Computational commutative algebra and operad theory

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Outline of the talk

- What is an operad?
- An example of a classification problem for operads.
- An example of a solution of a classification problem.

Operads (in algebra)

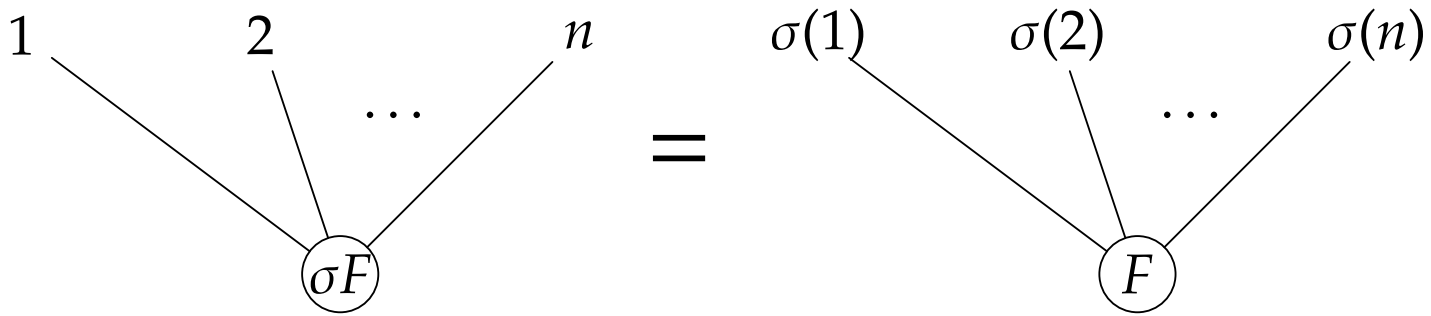
- To understand something about linear transformations of a vector space, it is often natural to formulate and answer questions in terms of associative algebras and their categories of modules.
- For operations with several arguments, a very convenient language is given by operads.
- Informally, for all algebras of some type, there exists one "higher algebra" (an operad) for which all these algebras are modules.

Permutations

Given an operation, we can permute its arguments:

$$(\sigma F)(x_1, \ldots, x_n) = F(x_{\sigma(1)}, \ldots, x_{\sigma(n)}),$$

or



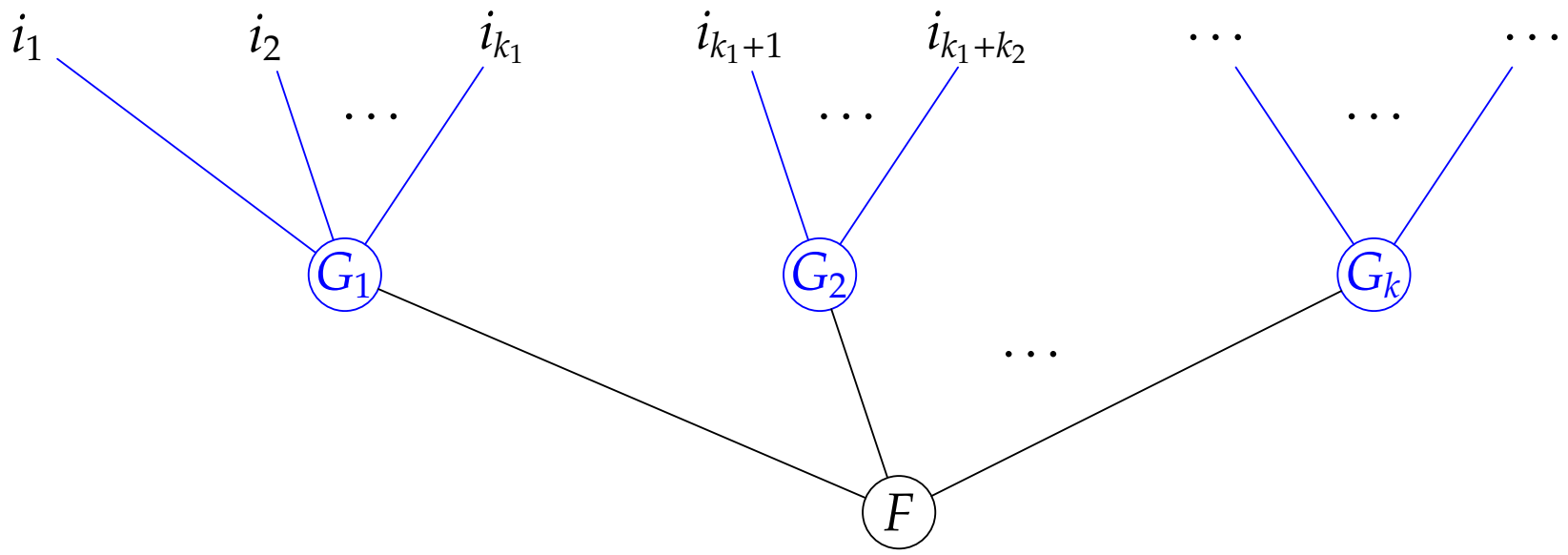
so that the set of all n-ary operations for a given algebraic structure is a representation of the symmetric group S_n.

Compositions

Also, we can compose operations:

$$\gamma_I(F, G_1, G_2, \ldots, G_k)(x_1, \ldots, x_n) =$$
$$= F(G_1(x_{i_1}, x_{i_2}, \ldots, i_{n_1}), G_2(x_{i_{n_1+1}}, \ldots, x_{i_{n_1+n_2}}), \ldots),$$

or



(note that there are composition maps γ_I for all sequences I, so in fact compositions incorporate permutations!)

Symmetric operads

"Definition": a (symmetric) operad is an abstraction of those structures (compositions and permutations).

The category of operads has free objects. Similar to how free algebras are spanned by monomials, free operads are spanned by terms, or by tree monomials.

We can talk about presenting operads via generators and relations. When a presentation has parameters, and we would like to learn something about how properties of an operad depend on parameters, computational commutative algebra enters the game.

Example: the operad *Lie*

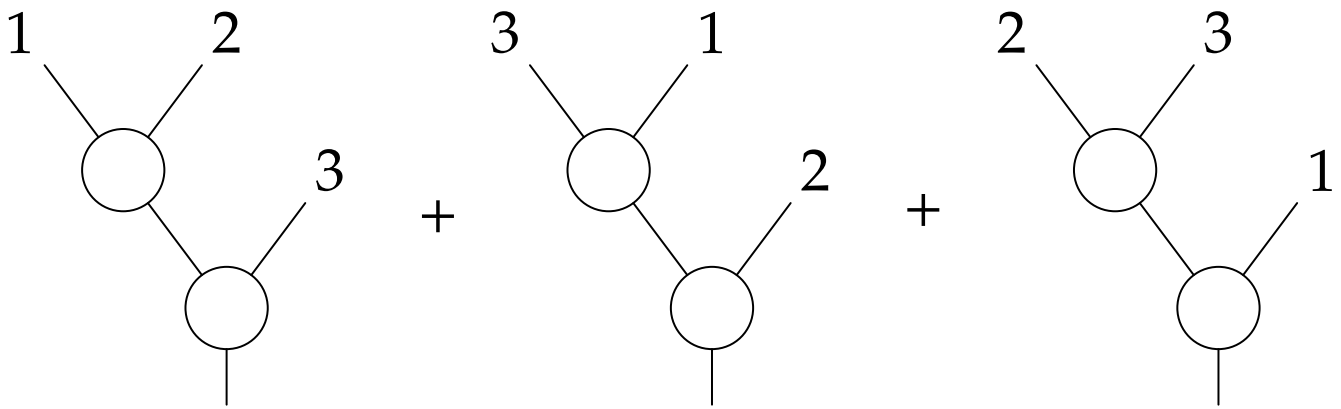
A naive definition. The operad *Lie* of Lie algebras consists of all Lie operations:

$$a_1 \mapsto a_1,$$
$$a_1, a_2 \mapsto [a_1, a_2],$$
$$a_1, a_2, a_3 \mapsto [[a_1, a_2], a_3], [[a_2, a_3], a_1], [[a_3, a_1], a_2],$$
$$\dots$$

Note that:
– all other operations you can think of can be expressed in terms of these, e.g. $[a_2, [a_1, a_3]] = [[a_3, a_1], a_2]$;
– the three operations in the third line are linearly dependent (Jacobi identity), thus the space of Lie operations with three arguments is two-dimensional.

Example: the operad *Lie*

A formal definition. The operad *Lie* of Lie algebras is the quotient of the free operad with one skew-symmetric binary generator modulo the two-sided ideal generated by the element



Thus, we take a skew-symmetric binary operation, and quotient out all algebraic consequences of the Jacobi identity.

Example of a question

Loday's parametrised one-relation operads: one generator $a_1, a_2 \mapsto a_1 a_2$, one relation

$$(a_1 a_2) a_3 = \sum_{\sigma \in S_3} x_\sigma \, a_{\sigma(1)} \big(a_{\sigma(2)} a_{\sigma(3)} \big) \ ;$$

Important instances of operads from this family are:

the associative operad, $(a_1 a_2) a_3 = a_1 (a_2 a_3)$,
the Leibniz operad, $(a_1 a_2) a_3 = a_1 (a_2 a_3) - a_2 (a_1 a_3)$,
the Zinbiel (dual Leibniz) operad, $(a_1 a_2) a_3 = a_1 (a_2 a_3) + a_1 (a_3 a_2)$,
the Poisson operad,
$(a_1 a_2) a_3 = a_1 (a_2 a_3) + \frac{1}{3} \big[a_1 (a_3 a_2) - a_2 (a_1 a_3) + a_2 (a_3 a_1) - a_3 (a_1 a_2) \big]$

Example of a question

For each of these operads, the space of n-ary operations is the regular representation of S_n. Equivalently, the underlying vector space of the free algebra generated by V is $T(V)$.

Question: classify all such "regular" operads.

Main theorem: over a quadratically closed fields, there are, up to isomorphism, five regular operads: associative, Poisson, Leibniz, Zinbiel, and the "left-nilpotent" operad $(a_1 a_2)a_3 = 0$.

Strategy: classify all answers using computational commutative algebra, and then use operad automorphisms and more computational commutative algebra to complete the classification.

Outline of the solution

Generate all consequences of arity 4; there are 120 of them. There are also 120 tree monomials of arity 4, so one expects the operad to collapse for generic parameters. (This is true: it collapses in the special case $(a_1a_2)a_3 = -a_1(a_2a_3)$, so collapses generically.)

Use representation theory of symmetric group: the 120-dimensional space of consequences generically looks as 5 times the regular module, and we want it to be 4 times the regular module. Split into a direct sum of isotypic components explicitly (Clifton's algorithm), and try to use determinantal ideals to impose conditions on ranks.

Determinantal ideals: concretely, impose the condition that the rank for the isotypic component indexed by a partition λ is at most k_λ for all λ (sum of determinantal ideals), and then remove solutions for which the rank for λ is at most $k_\lambda - 1$ for some λ. In our case, we deal with matrices of sizes $5, 15, 10, 15, 5$ which we want to be of ranks $4, 12, 8, 12, 4$ respectively. This altogether requires computing about 500,000 minors of various sizes (up to 13)

Outline of the solution

Pre-processing: partial Smith normal form (use constant matrix elements to reduce). This makes us deal with determinantal ideals of matrices of size $2, 5, 4, 5, 2$, and we want the ranks of those matrices to be $1, 2, 2, 2, 1$ respectively. This altogether requires computing less than 500 minors of various sizes (up to 3).

In comparison, if we merely wanted to impose the dimension condition in arity 4, then even after computing the partial Smith form, we would need to compute more than 10^{18} minors of size 12 and 13 of a matrix of size 36, and then compute Gröbner bases for the respective ideals. This is extremely far from feasible, plus there are many (families of) solutions that give the wrong S_4-representation, and careful sieving through solutions is also problematic.

Outline of the solution

Practicalities: when computing with `Maple`, we used the version which had known bugs in computing radicals of ideals. For that reason, we find it convenient to separate into several cases depending on which parameters are non-zero, so that one can simplify the equations one needs to solve. (In other words, very elaborate divide-and-conquer.)

If working with `Magma`, this issue does not arise, and even the capacity of the online calculator is enough to compute radicals of all the determinantal ideals in question.

Outline of the solution

Theorem

The parametrized one-relation operads with the regular module in arity 4 *are precisely the operads from the following list:*

(I) $(a_1a_2)a_3 = s\, a_3(a_1a_2)$,

(II) $(a_1a_2)a_3 =$
$a_1(a_2a_3) + s\big[a_1(a_3a_2) - a_2(a_1a_3) + a_2(a_3a_1) - a_3(a_1a_2)\big]$,

(III) $(a_1a_2)a_3 = u\big[a_1(a_2a_3) + a_1(a_3a_2) + a_3(a_2a_1)\big] - v\big[a_2(a_1a_3) + a_2(a_3a_1) + a_3(a_1a_2)\big] - a_3(a_2a_1)$,

(IV) $(a_1a_2)a_3 = u\big[a_1(a_2a_3) - a_2(a_1a_3) - a_3(a_2a_1)\big] + v\big[a_1(a_3a_2) - a_2(a_3a_1) - a_3(a_1a_2)\big] + a_3(a_2a_1)$,

where in the case (i) we require $s \neq \pm 1$, *in the case (ii) we require* $s \neq -1$, *and in both cases (iii) and (iv) the point* (u, v) *belongs to the hyperbola* $y^2 - y - (x-1)^2 = 0$ *with the points* $(1, 1)$ *and* $(\frac{1}{3}, -\frac{1}{3})$ *excluded.*

Outline of the solution

Change of variables: $a_1 a_2 = a_1 \star a_2 + t\, a_2 \star a_1$.

Idea: perform this change of variables, and find a value of t for which (a) the result is a parametrised one-relation operad (some constraints $t \neq C$) and (b) that operad is one of our five operads (trivial, associative, Poisson, Leibniz, Zinbiel). The "bad" values of t are obtained by computing some resultants / Gröbner bases: computational commutative algebra again!

Outline of the solution

For the family $(a_1a_2)a_3 = s\, a_3(a_1a_2)$, can find t to change it into $(a_1 \star a_2) \star a_3 = 0$.

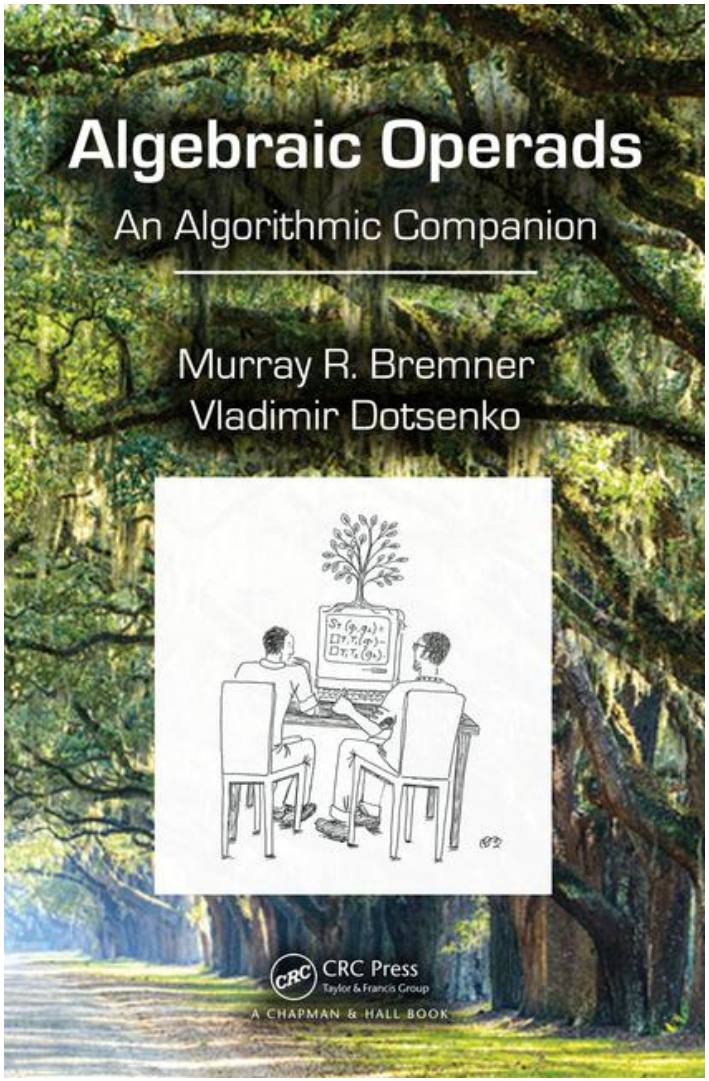
For the family
$(a_1a_2)a_3 = a_1(a_2a_3) + s\big[a_1(a_3a_2) - a_2(a_1a_3) + a_2(a_3a_1) - a_3(a_1a_2)\big]$,
can find t changing it into the associative operad for $s \neq 1/3$; for $s = 1/3$ the solution is stable under the change of variable, and is the Poisson operad.

For the families
$(a_1a_2)a_3 = u\big[a_1(a_2a_3) + a_1(a_3a_2) + a_3(a_2a_1)\big] - v\big[a_2(a_1a_3) + a_2(a_3a_1) + a_3(a_1a_2)\big] - a_3(a_2a_1)$,
$(a_1a_2)a_3 = u\big[a_1(a_2a_3) - a_2(a_1a_3) - a_3(a_2a_1)\big] + v\big[a_1(a_3a_2) - a_2(a_3a_1) - a_3(a_1a_2)\big] + a_3(a_2a_1)$,
can find t changing them into Zinbiel and Leibniz, respectively. (Hint: $v^2 - v - (u-1)^2 = 0$, so forcing $v = 0$ automatically guarantees $u = 1$.)

Concluding remarks

Gröbner bases for operads (D–Khoroshkin 2010) are not applicable to this problem, since the crucial idea behind Gröbner bases for operads is to use "shuffle operads" and forget as much about symmetric groups actions as at all possible.

A huge miracle: classifying operads that are regular up to arity 4 leads to a complete classification of regular operads. In a way, this should be the manifestation of a (yet unknown) version of Drinfeld's "Koszul deformation principle" in the case of operads.

Thank you for your attention!

For more information about operads: